From Mentions to Modularity: Co-Occurrence Networks in Ancient Greek and Latin Texts

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Abstract

Network analysis of individuals mentioned in historical texts offers a powerful means of exploring large corpora, encompassing both factual and fictional narratives. Previous work has demonstrated that manually constructed social networks from antiquity, based on interactions and relations in sources such as the *Iliad*, the *Odyssey*, and the *Pentateuch,* exhibit structural properties comparable to real-world social networks (Mac Carron and Kenna 2012; Massey 2016; Miranda et al. 2018). Other studies suggest that co-occurrence networks, while less precise, can approximate the results of carefully curated networks (Dörpinghaus 2022; Hutchinson and and Louwerse 2018). Building on this foundation, the present study applies a co-occurrence approach to ancient texts in order to assess how accurately such networks capture known features of these works.
Our analysis focuses on clustering patterns in a co-occurrence network of individuals drawn from a corpus of Ancient Greek and Latin texts, with a particular focus on texts mentioning Plato. Co-occurrence is defined using a sliding window of two sentences, following the method of Smeets (2021). Edge weights are calculated per work with a formula based on Salton's cosine (Hamers et al. 1989). For the network of all works combined, the mean of the weight over all works is used.

We compare 50 iterations of two different modularity-based clustering methods for the largest connected component of the network:

the Louvain algorithm (Blondel et al. 2008), based on modularity optimization, implemented with networkx louvain_communities

the Leiden algorithm (Traag et al. 2019) implemented with leidenalg

This comparison enables inspection of cluster stability and selection of the appropriate algorithm. While Leiden achieves slightly higher modularity than Louvain, the difference alone is not sufficient. Using TF–IDF-derived keywords, we find that Leiden produces more stable top-5 keyword sets across runs, with a larger proportion of clusters retaining top keywords in over half of the runs. The Plato-centered cluster ('dionysios', 'platon', 'platons', 'syrakus', 'sohn') remains highly stable across both methods (45 Louvain vs. 40 Leiden runs), indicating that strongly defined communities are largely unaffected by algorithm choice. Differences instead in medium and smaller clusters, where Leiden produces more philologically meaningful groupings, whereas larger clusters remain largely comparable, though typically reduced

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in size. Overall, this suggests that Leiden yields more robust and semantically coherent partitions. We therefore adopt Leiden with POTTS_GAMMA = 0.6, which is, as shown in the left graph of Figure 1, where the modularity value plateaus.

For this clustering, we further investigate whether nodes with shared attributes, such as cultural background, or philosophical affiliation, tend to cluster together. For this, we use two methods: first, general person attributes are extracted from Wikidata to see if clustering relates to associated sex/gender, time periods, associated regions, or occupation, second, we employ the already mentioned qualitative inspection of the coherency within descriptive texts for the individual.

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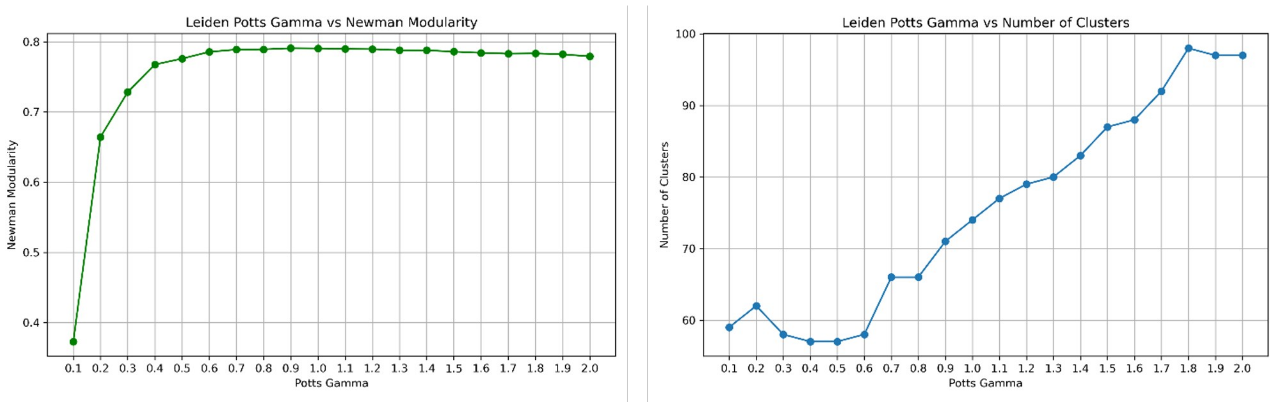

Figure 1.

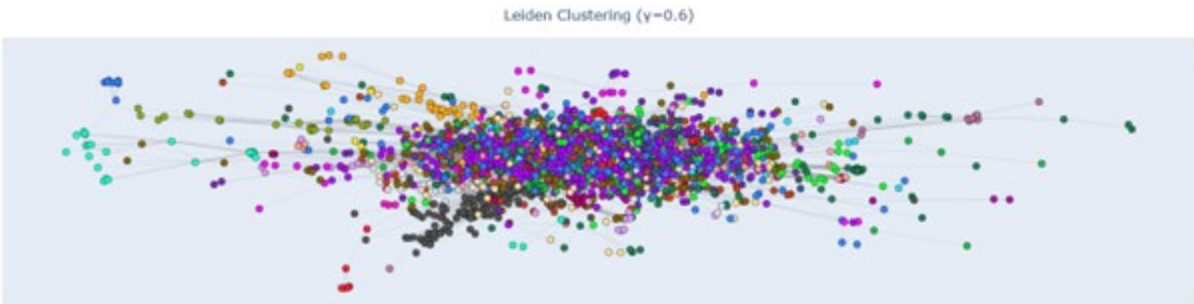

Figure 2. Visualisation of the largest connected component in the network. Clustered with the Leiden algorithm.